04. Agriculture

01. What is agriculture?

Science and the art of cultivation of crops and rearing of animals (life stocks).

02. What is the importance of agriculture?

- It provides food
- The agro-based industries depend on agriculture for raw materials
- It provides employment to rural population
- Agriculture is the base of our economy as our economy is agro-based
- It provides us with animal products also
- We earn foreign exchange by exporting agricultural products.

Types of Farming
- Subsistence
  - Primitive
  - Intensive Grain Farming
- Commercial
  - Extensive Grain Farming
  - Plantation

Primitive –

- In this type of farming, primitive tools are used for cultivation.
- The cultivators grow common varieties of crops of vegetables, paddy, maize, and few millets.
- Family members work in the fields and modern technology is not in use.
- Crops are grown in small quantities to fulfill the needs of families and surplus product is sold in the local weekly market (termed as 'haat').
- The cultivators depend on rainfall for watering their crops.
- The cultivators clear a patch of the forest by cutting down of the tress and burning them on the land as they believe that the ashes increase the fertility of the land. That is why, it is also called as 'slash and burn' cultivation.
- They continue to grow the same crop for 5-6 years without adding any extra imports like fertilizers, pesticides, etc. This decreases the fertility of the soil. So, the cultivators abandon the fields and shift to the next part of the forest to continue the process. This is also called as 'shifting cultivation'.

03. Differentiate between intensive grain farming and extensive grain farming.

Intensive	Extensive
This type of cultivation is practiced in densely populated regions.	This type of cultivation is practiced in moderately to sparsely populated regions.
The land holdings are small in size.	The land holdings are large in size.
It is highly labor intensive and very less machines are used.	Machines are used for every work.
HYV seeds of low value are grown.	HYV seeds of high value are grown
The farmers produce grains mainly for self-consumption and sell the surplus in the market.	The farmers grow grains mainly for selling purpose.

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Example – Paddy (rice) cultivation in West Bengal.	Example – Paddy (rice) cultivation in Punjab.
In both the types of farming, farmers use a large amount of pesticides, fertilizers, etc.	

Plantation –

- This type of farming is done on very large lands termed as "plantations".
- This is a capital-intensive farming.
- This is a labor-intensive farming.
- The plantations are well connected with means of transport and communication.
- A single crop is grown in this type of farming.
- In plantations, industrial crops are grown.
- Example – Bamboo, coconut, coffee, tea, banana, etc.

Cropping Pattern –

01. Rabi Crop –
    [i] Season of growth: Rabi crops are sown in winter from October to December and harvested in summer from April to June.
    [ii] Crops grown: Wheat, barley, peas, gram and mustard.
    [iii] Region of production: These crops are grown in large parts of India, states from the north and north-western parts such as Punjab, Haryana, Himachal Pradesh, Jammu and Kashmir, Uttarakhand and Uttar Pradesh.
    [iv] Reason of production:
    - Availability of precipitation during winter months due to the western temperate cyclones helps in the success of these crops.
    - The success of the green revolution.
02. Kharif Crop –
    [i] Season of growth: Kharif crops are grown with the onset of monsoon in different parts of the country and these are harvested in September-October.
    [ii] Crops grown: Paddy, maize, jowar, bajra, tur (arhar), moong, urad, cotton, jute, groundnut and soyabean.
    [iii] Region of production: The most important rice-growing regions are Assam, West Bengal, coastal regions of Odisha, Andhra Pradesh, Telangana, Tamil Nadu, Kerala and Maharashtra, particularly the (Konkan coast) along with Uttar Pradesh, Bihar, Punjab and Haryana.
    [iv] Reason of production: Rainfall caused due to monsoon winds.
03. Zaid Crop –
    [i] Season of growth: In between the rabi and the kharif seasons, there is a short season during the summer months known as the Zaid season.
    [ii] Crops grown: the crops produced during 'zaid' are watermelon, muskmelon, cucumber, vegetables, fodder crops and sugarcane.
    [iii] Region of production: None.
    [iv] Reason of production: Dry climatic conditions.

Major Crops –

04. Agriculture

01. Rice –

[i] Geographical requirements:

- Temperature – Requires high temperature, (above 25°C)
- Rainfall – Annual rainfall above 100 cm. In the areas of less rainfall, it grows with the help of irrigation
- Soil – Fertile Clayey Soil

[ii] Region of growth:

- Rice is grown in the plains of north and north-eastern India, coastal areas and the deltaic regions.
- Punjab, Haryana and western Uttar Pradesh and parts of Rajasthan.

[iii] Other characteristic features:

- It is the staple food crop of a majority of the people in India. Our country is the second largest producer of rice in the world after China.
- Development of dense network of canal irrigation and tubewells have made it possible to grow rice in areas of less rainfall.

02. Wheat –

[i] Geographical requirements:

- Temperature – This rabi crop requires a cool growing season and a bright sunshine at the time of ripening
- Rainfall – It requires 50 to 75 cm of annual rainfall
- Soil – Well fertile loamy soil

[ii] Region of growth:

- There are two important wheat-growing zones in the country – the Ganga-Satluj plains in the north-west and black soil region of the Deccan.
- The major wheat-producing states are Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Bihar and Rajasthan.

[iii] Other characteristic features:

- This is the second most important cereal crop. It is the main food crop, in north and north-western part of the country.

03. Millets –

[i] Geographical requirements:

- Temperature – High
- Rainfall – It is a rain-fed crop mostly grown in the moist areas which hardly needs irrigation
- Soil – (i) Bajra grows well on sandy soils and shallow black soil.
  (ii) Ragi is a crop of dry regions and grows well on red, black, sandy, loamy and shallow black soils

[ii] Region of growth:

- Major Jowar producing States are Maharashtra, Karnataka, Andhra Pradesh and Madhya Pradesh.
- Major Bajra producing States are Rajasthan, Uttar Pradesh, Maharashtra, Gujarat and Haryana.
- Major ragi producing states are: Karnataka, Tamil Nadu, Himachal Pradesh, Uttarakhand, Sikkim, Jharkhand and Arunachal Pradesh.

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[iii] Other characteristics features:
- Jowar, bajra and ragi are the important millets grown in India.
- Though, these are known as coarse grains, they have very high nutritional value. For example, ragi is very rich in iron, calcium, other micro nutrients and roughage.

04.Maize –

[i] Geographical requirements:
- Temperature – Requires temperature between 21°C to 27°C
- Rainfall – Moderate rainfall
- Soil – Grows well in old alluvial soil

[ii] Region of growth:
- Major maize-producing states are Karnataka, Madhya Pradesh, Uttar Pradesh, Bihar, Andhra Pradesh and Telangana.

[iii] Other characteristic features:
- It is a crop which is used both as food and fodder.
- In some states like Bihar maize is grown in rabi season also.
- Use of modern inputs such as HYV seeds, fertilisers and irrigation have contributed to the increasing production of maize.

05.Pulses –

[i] Geographical requirements:
- Temperature – Some grow in Rabi season and some in Kharif season
- Rainfall – Pulses need less moisture and survive even in dry conditions
- Soil – Variety of well fertile soil

[ii] Region of growth:
- Major pulse producing states in India are Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh and Karnataka.

[iii] Other characteristic features:
- India is the largest producer as well as the consumer of pulses in the world. These are the major source of protein in a vegetarian diet.
- Major pulses that are grown in India are tur (arhar), urad, moong, masur, peas and gram.
- Being leguminous crops, all these crops except arhar help in restoring soil fertility by fixing nitrogen from the air. Therefore, these are mostly grown in rotation with other crops.

Food crops other than grains –

01. Sugarcane –

[i] Geographical requirements:
- Temperature – From 21°C to 27°C
- Rainfall – An annual rainfall between 75 cm and 100 cm. Irrigation is required in the regions of low rainfall
- Soil – It can be grown on a variety of fertile soils

[ii] Region of growth: The major sugarcane-producing states are Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Telangana, Bihar, Punjab and Haryana.

[iii] Other characteristic feature:
- Needs manual labour from sowing to harvesting.

- India is the second largest producer of sugarcane only after Brazil.
- It is the main source of sugar, gur (jaggary), khandsari and molasses.

02. Oil Seeds –

[i] Geographical requirements:

- Temperature – Some grow in Rabi season and some in Kharif season
- Soil – Variety of fertile soil

[ii] Region of growth: Gujarat was the largest producer of groundnut followed by Rajasthan and Tamil Nadu in 2019–20.

[iii] Other characteristic features:

- In 2020 India was the second largest producer of groundnut in the world after China.
- Different oil seeds are grown covering approximately 12 per cent of the total cropped area of the country.
- Main oil-seeds produced in India are groundnut, mustard, coconut, sesamum (til), soyabean, castor seeds, cotton seeds, linseed and sunflower.
- Most of these are edible and used as cooking mediums.
- Some of these are also used as raw material in the production of soap, cosmetics and ointments.
- Groundnut is a kharif crop and accounts for about half of the major oilseeds produced in the country.
- Linseed and mustard are rabi crops. Sesamum is a kharif crop in north and rabi crop in south India.
- Castor seed is grown both as rabi and kharif crop.

03. Tea –

[i] Geographical requirements:

- Temperature – The tea plants grow well in tropical and sub-tropical climates
- Rainfall – Tea bushes require warm and moist frost-free climate all through the year. Frequent showers evenly distributed over the year ensure continuous growth of tender leaves
- Soil - Deep and fertile well-drained soil, rich in humus and organic matter

[ii] Region of growth: Major tea-producing states are Assam, hills of Darjeeling and Jalpaiguri districts of West Bengal, Tamil Nadu and Kerala, Himachal Pradesh, Uttarakhand, Meghalaya, Andhra Pradesh and Tripura

[iii] Other characteristic features:

- Tea cultivation is an example of plantation agriculture.
- It is also an important beverage crop introduced in India initially by the British.
- Tea is a labour-intensive industry.
- It requires abundant, cheap and skilled labour.
- Tea is processed within the tea garden to restore its freshness.

04. Coffee –

[i] Geographical requirements:

- Temperature – High
- Rainfall – High
- Soil – Deep and well fertile soil

[ii] Region of growth: Initially its cultivation was introduced on the Baba Budan Hills and even today its cultivation is confined to the Nilgiri in Karnataka, Kerala and Tamil Nadu.

[iii] Other characteristic features:

- Indian coffee is known in the world for its good quality.
- The Arabica variety initially brought from Yemen is produced in the country. This variety is in great demand all over the world.

Non-Food Crops –

01. Rubber –

[i] Geographical requirements:

- Temperature – Above 25°C
- Rainfall – More than 200 cm
- Soil – Well fertile soil

[ii] Region of growth: It is mainly grown in Kerala, Tamil Nadu, Karnataka and Andaman and Nicobar Islands and Garo hills of Meghalaya.

[iii] Other characteristic features: It is an equatorial crop, but under special conditions, it is also grown in tropical and sub-tropical areas.

02. Cotton –

[i] Geographical requirements:

- Temperature – It requires high temperature, 210 frost-free days and bright sun-shine for its growth. It is a kharif crop and requires 6 to 8 months to mature
- Rainfall – Light rainfall or irrigation
- Soil – Variety of fertile soils, grows best in black cotton soil

[ii] Region of growth: Major cotton-producing states are– Maharashtra, Gujarat, Madhya Pradesh, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Punjab, Haryana and Uttar Pradesh.

[iii] Other characteristic features:

- India is believed to be the original home of the cotton plant. Cotton is one of the main raw materials for cotton textile industry.
- India is second largest producer of cotton after China.

03. Jute –

[i] Geographical requirements:

- Temperature – High temperature is required during the time of growth
- Rainfall – High, above 100 cm
- Soil – Fertile clayey soil

[ii] Region of growth: West Bengal, Bihar, Assam, Odisha and Meghalaya are the major jute producing states.

[iii] Other characteristic features:

- It is known as the golden fibre.
- It is used in making gunny bags, mats, ropes, yarn, carpets and other artefacts.

Technological and Institutional Reforms –

01. Technological –

[i] The Green Revolution based on the use of package technology

[ii] The White Revolution (Operation Flood)
[iii] Special weather bulletins
[iv] Agricultural programmes for farmers were introduced on the radio and television.
[v] Soil testing centres
[vi] Mobile apps to provide solutions to problems related to crops

02. Institutional –

[i] Rejection of Collectivisation and consolidation of land holdings
[ii] Cooperation and abolition of zamindari
[iii] Provision for crop insurance against drought, flood, cyclone, fire and disease
[iv] Establishment of Grameen banks, cooperative societies and banks for providing loan facilities to the farmers at lower rates of interest.
[v] Kisan Credit Card (KCC), Personal Accident Insurance Scheme (PAIS) are some other schemes introduced by the Government of India for the benefit of the farmers.
[vi] The government also announces minimum support price, remunerative and procurement prices for important crops to check the exploitation of farmers by speculators and middlemen.
[vii] Agricultural universities have been opened in every district of India.
[viii] Veterinary hospitals have been opened in every district of India.

Sources: NCERT – Contemporary India-II Textbook

Shibani Banerjee Ma'am – Geography teacher, BMSSS

- Aayan